

Attitudes towards political violence in Chile’s cycle of political conflict: A longitudinal analysis focusing on education and participation

René Canales-Sellés¹, Daniel Miranda²

¹Bremen International Graduate School of Social Sciences, University of Bremen

²Department of Psychology, University of Chile

June, 2026

Abstract

The Chilean social unrest of October 2019 poses a paradox for political theory: Why did highly educated sectors—typically associated with democratic moderation—legitimize forms of disruptive violence? Using longitudinal data from ELSOC (2016–2023; 20,007 observations, 3,666 individuals), this article examines the relationship between educational attainment, participation in protests, and the justification of violence through multilevel models. The results challenge the “civilizing effect” of education: among non-participants, higher education reduces the justification of violence; among protesters, it significantly increases it. Mobilized university students exhibit the highest levels, reversing the expected effect. This pattern reflects a moral duality: educated participants reject state violence while legitimizing protest violence, thereby developing differentiated normative frameworks. These findings explain how the cross-sectional nature of the Chilean movement emerged from the rearticulation of moral frameworks across professional sectors, with implications for the study of nonviolent radicalization, political socialization, and the legitimacy of transgression in democracies with structural inequality.

Keywords: Protests · Violence · Stratification · Attitudes · Chile

1 Introducción

October 18, 2019, marked a turning point in Chile’s political history. What began as a mass fare evasion in Santiago’s metro system—sparked by students in response to rising transit fares—quickly evolved, within a matter of hours, into the most significant protest since the return to democracy (Somma et al., 2021). For several weeks, thousands of people took to the streets across the country, challenging both the neoliberal legacy of the dictatorship (Canales Cerón, 2022; Garcés, 2019; Araujo, 2022) and the historical limits of collective action (Somma et al., 2021; Cox et al., 2024; Pleyers, 2023). The so-called “Chilean Spring” combined mass participation, social transversality, and confrontational tactics in unprecedented proportions: from massive marches to road blockades, clashes with the Carabineros, and attacks on public infrastructure (Rivera et al., 2025).

The government’s response was equally extraordinary. The administration of then-President Sebastián Piñera declared a state of emergency and deployed military personnel to the streets for the first time since the restoration of democracy (Somma et al., 2021; Thaler et al., 2023). The crackdown resulted in more than 8,000 victims of police violence, over 400 cases of eye trauma, and at least 34 deaths, constituting—according to Amnesty International—the most serious human rights crisis since the end of the dictatorship (Amnistía, 2020; Human, 2019; Rodríguez et al., 2021). This scenario intensified media and political polarization and brought a persistent question to the center of public debate: when, and for whom, is political violence legitimate?

Both forms of violence are driven by different logics and have distinct effects on protest dynamics. State violence refers to

repressive actions carried out by state agents—such as the police or the armed forces—with the aim of dispersing or demobilizing demonstrators, a phenomenon some scholars refer to as “protest policing” (Steinert-Threlkeld et al., 2022). Protester violence, by contrast, refers to disruptive or destructive physical actions carried out by protesters themselves during a mobilization. Beyond their origins, the two forms also differ in their effects: while state violence can produce either mobilization or demobilization depending on its intensity, protester violence tends to consistently reduce the size of protests by eroding their legitimacy and discouraging third-party participation (Steinert-Threlkeld et al., 2022).

Several studies have shown that the justification of violence during the uprising was not uniform. Disi Pavlic et al. (Disi Pavlic et al., 2025) show that its acceptance varies according to temporal, spatial, and ideological proximity, while González and Le Foulon (Gonzalez et al., 2020) find that the most active protesters—young, educated, and left-leaning individuals—justified protest violence more frequently than those on the margins of these events. This pattern is counterintuitive given the classical literature associating education with greater acceptance of democratic norms and a rejection of violence (Dewey et al., 2024; Lipset, 1959; Nie et al., 1996). How, then, can we explain that the most educated sectors showed a greater willingness to justify violence during the mobilization cycle?

This assumption has clear theoretical roots. The resource model (Brady et al., 1995; Verba et al., 2002) argues that political participation and attitudes depend on differential access to resources—education, time, money, and civic skills—such that higher levels of education are associated with more conventional attitudes and greater commitment to institutional norms. However, much of the literature on political attitudes and conflict has extended this assumption beyond its original conditions, treating the effects of resources such as education as contingent on the context of action and individuals’ participatory experiences. McAdam (D. McAdam et al., 2009) demonstrated that high-risk activism has lasting “biographical consequences”: it alters interpretive frameworks, collective identities, and moral assessments of the legitimacy of action. Participating in protests exposes individuals to alternative discourses, fosters group solidarity, and, in repressive contexts, can directly confront them with state violence (Snow et al., 1988; Olcese et al., 2014). These experiences not only reshape political attitudes but also influence how the effects of education are expressed in the justification of violence (Dahlum, 2019). Participation in protest therefore does not uniformly influence attitudes; rather, it activates differential potentials for change rooted in biographical trajectories and repertoires of collective action (Doug McAdam, 1989; Tarrow, 1998; Snow et al., 1988). In this sense, the justification of political violence constitutes a central dimension for the health of democracies: when broad sectors of the citizenry perceive violence as a legitimate resource for collective action, the normative foundations of the democratic order are eroded and the space for radicalization and conflict expands (Muller et al., 1990; Norris, 2011). Understanding which factors shape these attitudes—and under what conditions they change—is therefore essential to understanding the stability and quality of contemporary political systems.

This study proposes an alternative model for understanding attitudes toward political violence. The central argument is that the effects of education on such attitudes depend on participation in protests and the type of violence under consideration. Education does not function as a linear and universal predictor, but rather as a cognitive amplifier whose effect is conditioned by participatory experience: among those who do not participate, it reinforces the rejection of violence; among those who do, it provides more elaborate interpretive frameworks to legitimize it in specific contexts of mobilization. This logic is further expressed in a normative distinction between types of violence: more educated and mobilized individuals tend to justify protest violence while rejecting state violence, a pattern consistent with theories of legitimacy and procedural justice (Davenport, 1995;

Muñoz et al., 2019; Tyler, 2006).

It is important to recognize, however, that this is not a static phenomenon. Attitudes toward political violence vary over time depending on the intensity of the conflict cycle: the levels and forms of justification observed before, during, and after the outbreak differ substantially, producing attitudinal trajectories that reconfigure themselves as the mobilization context evolves (Disi Pavlic, 2021). Addressing this temporal dimension is therefore not a secondary methodological refinement, but a necessary condition for capturing the dynamic nature of the phenomenon. The guiding question of this study is consequently: how do education and participation in protests interact to shape attitudes toward political violence in the Chilean context over time? This study aims to contribute to two bodies of literature. First, to the literature on education and political attitudes, by assessing the “civilizing effect” as contingent upon participatory context (Nie et al., 1996; Weakliem, 2002). Second, to the literature on political participation and socialization (Doug McAdam, 1989; Giugni, 2004), by examining whether participation interacts with cognitive (educational) resources in complex ways, thereby shaping individuals’ interpretive frameworks. Taken together, this study seeks to shed light on the foundations of the legitimization of violence during Chile’s period of political conflict, examining how participation in protests can transform moral frameworks and enable the justification of certain forms of normative transgression as an expression of political conflict. The Chilean case offers an exceptional empirical setting for this purpose. The 2019 social uprising not only triggered massive participatory dynamics but also condensed, within a brief period, an unusual combination of structural discontent, institutional exhaustion, and state repression perceived as illegitimate—conditions that together allow for a precise examination of how interpretive resources and participatory experiences interact in shaping attitudes toward political violence. Analyzing this contentious cycle thus allows us to empirically test a more general argument: that the subjective legitimacy of political violence is not a direct reflection of stable individual attributes, but rather the result of a contingent interaction between cognitive resources and experiences of mobilization.

2 Theoretical Background

2.1 Justification of Violence

The willingness to justify violence in protest contexts is a complex phenomenon, one that distinguishes between attitudes toward violence exercised by protesters and that employed by state forces (Marx, 1998; Disi Pavlic et al., 2025). Protester violence tends to be evaluated as destabilizing to the political order and therefore illegitimate by definition, while state violence—protest policing—is framed as a functional measure for maintaining social order (Davenport et al., 2011; ?), granting it a presumption of institutional justification that protester violence lacks (Disi Pavlic et al., 2025).

This normative asymmetry has empirical consequences: civic judgment does not operate neutrally, but rather against differentiated expectations depending on which actor exercises force (Jämte et al., 2024). Approval of police violence may increase when directed against unpopular groups (Ellner, 2003), but the same violence erodes state legitimacy when perceived as unjust (Górecki, 2013; Pérez-Liñán et al., 2010). Conversely, protester violence can shift from illegitimate to justifiable when the state is perceived as the original aggressor (Disi Pavlic et al., 2025).

2.2 Policing and Its Effects on Citizens' Attitudes

Research shows that policing style has substantive effects on how citizens evaluate violence in protests. Escalated force can trigger radicalization and escalation when perceived as disproportionate (Koopmans, 1993; Francisco, 1995), eroding the legitimacy of both the police and the government (Bonner et al., 2022). The literature distinguishes between a negotiated management model, which prioritizes communication and minimal use of force, and an escalated force model, based on coercion (Della Porta et al., 2007); and more recently, between active policing—direct intervention involving chemical agents, projectiles, and arrests—and passive policing—selective monitoring and containment (Disi Pavlic et al., 2025).

Drawing on procedural justice theory (Tyler et al., 2003), when police conduct is perceived as unjust, institutional legitimacy is eroded and the justification of resistance increases (Monica M. Gerber et al., 2017; Sunshine et al., 2003). Crucially, these effects are not uniform: they vary according to ideology and prior protest experience (Disi Pavlic et al., 2025). In Chile, the legitimacy crisis of the Carabineros during the 2019 uprising illustrates this mechanism: the widespread reports of human rights violations normalized the justification of protest violence as legitimate self-defense (Mónica M. Gerber et al., 2023; Carrasco Paillamilla et al., 2023). What is particularly relevant for this study is that direct experience with policing—through participation in protests—constitutes one of the primary triggers of attitudinal change.

2.3 Education and Attitudes toward Violence in Protests

The classical “civilizing effect” argument (Lipset, 1959; ALMOND et al., 1963) holds that education fosters the internalization of democratic norms and reduces the propensity to accept political violence, by developing cognitive skills that enable individuals to adopt moderate positions (Converse, 1972; Nie et al., 1996), promoting values of tolerance and institutional respect (Siegel, 1981), and strengthening subjective political efficacy (Dyrstad et al., 2020). Under this paradigm, the justification of violence is a residue of educational deficit.

However, this model faces significant empirical limitations. Studies with robust causal designs show that the direct effects of education on political attitudes are largely explained by self-selection (Kam et al., 2008; Berinsky et al., 2011). Moreover, counterintuitive evidence—such as the presence of highly educated individuals in violent organizations—suggests that the effects of education are context-contingent (Krueger et al., 2003; Fair et al., 2006).

This has prompted a reconceptualization: rather than a universal inhibitory effect, education operates as a conditional cognitive amplifier (Nie et al., 1996; Sabucedo et al., 2018). Its primary function is not to instill tolerance, but to provide cognitive sophistication—the analytical tools to construct complex moral frameworks that can both delegitimize and legitimize violence depending on situational context. In settings of high inequality and institutions perceived as unjust, this sophistication enables individuals to frame violence as an exceptional yet principled response consistent with notions of justice (della Porta, 2017; Varlık et al., 2025).

The key activation mechanism is direct participation in protests. The situated experience of protesting exposes individuals to state repression, to elaborated ideological narratives that distinguish between types of violence (Tilly, 2003; Benford et al., 2000), and to identity pressures that reconfigure the relative weight of different moral considerations (Polletta et al., 2001; Van Stekelenburg et al., 2013). More educated individuals possess greater capacity to manage these tensions through cognitive compartmentalization: they can maintain general democratic values while constructing situational exceptions that frame protest

violence as “legitimate self-defense” (Tetlock, 1986; Disi Pavlic et al., 2025). Less educated individuals, by contrast, tend toward more consistent positions that do not vary as much with context (Krueger et al., 2003).

In this way, the effect of education is conditional on protest experience: among non-participants, the civilizing effect operates by reducing the justification of violence; among participants, cognitive sophistication is activated to tactically legitimize what would otherwise be rejected in the abstract.

This gives rise to what may be termed (H1) the education paradox effect, whereby:

- H1a: Among individuals who do not participate in protests, higher educational attainment is associated with lower justification of violence in demonstrations, consistent with the “civilizing effect” of education.
- H1b: Among individuals who do participate in protests, the effect of education reverses: university-educated individuals show greater justification of violence in demonstrations compared to those with lower levels of education.

The explanation lies in the fact that education fosters more sophisticated political thinking, but this operates differently depending on participation. Non-participants draw on this capacity to reinforce their commitment to nonviolent democracy and abstract norms of tolerance. Participants, by contrast, employ that same sophistication to construct ideological frameworks that allow them to tactically legitimize violence as a valid political instrument in specific protest contexts, without necessarily abandoning their broader democratic values.

2.4 Protest Participation and Attitudinal Change

The classical literature on political participation has tended to treat it as the result of pre-existing characteristics: people participate because they already hold certain attitudes, resources, and predispositions (Verba et al., 2002). More recent research, however, has challenged this linear model, showing that participation not only reflects prior orientations but transforms them (Doug McAdam, 1989; Passy et al., 2018). McAdam’s seminal work (Doug McAdam, 1989) on the 1964 Freedom Summer project was pioneering in documenting this effect: comparing participants with a control group of similar characteristics who ultimately did not attend, he found that those who did participate underwent lasting transformations in their political and moral dispositions, regardless of their initial predispositions.

Since then, the literature has identified two main mechanisms through which participation produces attitudinal change (Passy et al., 2018; Crossley, 2003). The first is exposure to new cognitive frameworks: social movements elaborate interpretive schemas that allow for the reframing of conflict, justice, and violence (Snow et al., 1988; Benford et al., 2000). Participation embedded in these environments can produce frame transformation, whereby previously illegitimate practices—such as certain forms of violence—are recast as “legitimate self-defense” or “necessary resistance.” The second mechanism is direct experience with state repression, which generates a moral reframing: what was previously “illegitimate violence” becomes “defensive violence,” while police action ceases to be perceived as order and comes to be seen as structural violence (Tyler et al., 2003; Monica M. Gerber et al., 2017).

Both mechanisms operate asymmetrically with respect to the actor exercising violence. Exposure to repression tends to increase the justification of protester violence by validating narratives of resistance, while simultaneously reducing the justification of state violence by eroding its perceived legitimacy (Disi Pavlic et al., 2025; ?). Evidence from the Chilean case is consistent

with this pattern: participation in the 2019 social uprising increased acceptance of confrontational tactics among protesters, particularly among those who experienced intense forms of state coercion (Rivera et al., 2025; Disi Pavlic et al., 2025).

2.5 Heterogeneity by Education

This transformative effect is not homogeneous, but rather mediated by participants' social and educational position (Giugni, 2004; Passy et al., 2018). As argued in the previous section, individuals with higher education possess greater cognitive sophistication to process and rearticulate complex moral frameworks (Nie et al., 1996). In participatory contexts, this capacity is activated: more educated individuals have greater resources to internalize the justificatory frameworks circulating within social movements and to construct situational exceptions that legitimize protest violence without abandoning broader democratic values (Tetlock, 1986; Sabucedo et al., 2018). Less educated individuals, by contrast, tend toward more stable positions that do not vary as much in response to participatory experience (Krueger et al., 2003).

It should be noted, however, that these effects may be temporally unstable. Rivera et al. (Rivera et al., 2025) and Disi Pavlic et al. (Disi Pavlic et al., 2025) document that attitudinal changes induced by participation tend to attenuate over time, suggesting that biographical transformation depends not only on the participatory experience itself, but on the continuity of activism and the structural conditions that sustain it.

The central hypothesis is that the transformative effects of participation are not uniform, but rather depend on the interaction between educational socialization and protest experience. Education provides the cognitive capacity to construct sophisticated moral justifications; participation, in turn, provides the emotional and symbolic conditions that make such elaboration necessary. The result is a bidirectional reconfiguration of moral frameworks: those who participate reinterpret state violence as illegitimate and protester violence as legitimate. Given the foregoing, we expect:

- H2a: University-educated individuals who participate in protests show greater justification of violence exercised by protesters, compared to university-educated non-participants.
- H2b: The same participating university-educated individuals show lower justification of violence exercised by the state, compared to university-educated non-participants.

Participation in protests is thus understood not as a passive reflection of pre-existing attitudes, but as a process of moral and political relearning, in which more educated individuals—precisely because of their cognitive sophistication—are most capable of coherently reformulating their normative principles in light of experience. In this way, collective action becomes the space in which the tensions between education, structural inequality, and the legitimacy of political violence are rearticulated.

3 The Chilean Case: From Structural Discontent to Justification

3.1 The Breakdown of the Social Compact

The study of the Chilean protest cycle that began in October 2019 is critical not only because of its historical magnitude, but because it represents the collapse of the post-transition social compact and offers a critical laboratory for analyzing the relationship between inequality, education, and political violence. Chile did not experience a mere acceleration of prior trends,

but a tectonic rupture in democratic coexistence. According to data from the Observatorio de Conflictos (Somma et al., 2021), the final quarter of 2019 recorded more than 2,700 contentious events—a figure nine times higher than the historical average of conflict observed in the preceding decade.

This phenomenon is distinguished not only by its scale, but by the centrality that violence acquired as a repertoire of action and political expression. During the uprising, the proportion of protests involving violence exceeded 40%, a qualitative leap relative to previous cycles (Joignant et al., 2020; Joignant et al., 2024). Even more significant for the analysis of political attitudes is the absence of organic leadership that characterized the movement: unlike the 2011 student mobilizations (Joignant et al., 2020; Donoso et al., 2023), where actors and demands were clearly identified, in 2019 some 68% of contentious actions lacked an articulated or identifiable demand. This scenario of anomie and institutional overflow validates the need to study how citizens reconfigure their moral frameworks and their justification of violence when traditional channels of conflict resolution are perceived as exhausted. In this regard, the Chilean case illustrates how the triad of violence, poverty, and lack of educational opportunities—if not addressed in an integrated manner—generates cycles of structural inequality that perpetuate social conflict (Varlık et al., 2025).

3.2 The Legitimacy Crisis and the Dynamics of Intergroup Violence

To understand why violence became an acceptable resource for broad sectors of the population, it is essential to analyze the dynamic interaction between protesters and security forces. Political violence during this period was not a unidirectional phenomenon, but the result of a profound crisis of institutional legitimacy. Empirical evidence gathered during the conflict shows that the perception of procedural injustice in the conduct of the Carabineros—characterized by the excessive use of force and degrading treatment—drastically eroded the moral authority of the police (Mónica M. Gerber et al., 2023).

There is a significant inverse relationship between legitimacy and violence: the lower the perceived legitimacy of the police, the greater the justification of violence by protesters as a valid tool for “social change” (Mónica M. Gerber et al., 2023). This finding is crucial for the argument developed here, as it indicates that state repression, far from restoring order, generated a backlash effect that morally validated radicalization. By perceiving authority as unjust and illegitimate, citizens came to reframe civilian violence not as criminality, but as an act of resistance and the reclamation of rights. Chile thus configured itself during this period as a scenario of reciprocal intergroup violence, in which police violence for social control and civilian violence for political change mutually reinforced each other, breaking down the normative barriers that had traditionally contained conflict. The Educational Paradox: From the Promise of Mobility to the Politicization of Discontent

With respect to education, the Chilean case challenges classical modernization theories that linearly associate educational expansion with social pacification and democratic consolidation. While the country experienced a rapid massification of higher education—moving from a gross enrollment rate of 16.8% in 1990 to 59% in 2012—this process unfolded under a radical market logic that incubated new conflicts (Bellei et al., 2014). Enrollment expansion was not driven by the state, but by the proliferation of private institutions and financing mechanisms such as the State-Guaranteed Student Loan (CAE), resulting in 78% of higher education expenditure being borne by families—the highest share in the OECD (Bellei et al., 2014).

This structure generated a “welfare paradox”: rather than acting as a social equalizer, education reproduced stratification and imposed a heavy debt burden on working-class households and emerging middle sectors. Far from producing conformism,

this tension transformed students into critical political actors. As Donoso et al. (Donoso et al., 2023) argue, the 2011 student movement not only placed education on the public agenda but generated a “repoliticization” of citizenship, demonstrating that higher levels of education provide the cognitive and civic resources necessary to challenge the status quo. Education in Chile thus operated as a catalyst for mobilization, with the most educated sectors acting not as buffers against conflict, but as vanguards that articulated a structural critique of the neoliberal model.

4 Data and Methods

4.1 Data

This study uses data from the Longitudinal Social Study of Chile (ELSOC), a panel survey implemented by the Center for Social Conflict and Cohesion Studies (COES) that collects information on the country’s urban adult population. The survey is designed to examine individuals’ attitudes, perceptions, and behaviors regarding issues of social conflict and cohesion. Its panel design allows the same individuals to be tracked over time, offering a unique window into the evolution of Chilean society. The survey has been administered annually since 2016, with the exception of 2020 due to the COVID-19 pandemic, accumulating seven waves of data (2016, 2017, 2018, 2019, 2021, 2022, and 2023).

ELSOC employs a probabilistic, multi-stage cluster sampling design covering both major urban centers and medium and small cities across the country. The sampling frame was stratified proportionally by urban population size (in six categories), with households randomly selected from 1,067 census blocks. The target population includes men and women aged 18 to 75 residing in private dwellings.

For the analyses reported here, and after applying an inclusion criterion requiring respondents to have participated in at least three waves, the analytical sample comprises 20,007 observations nested within 3,666 individuals. The survey, its methodological manuals, and datasets are publicly available through the COES portal <https://coes.cl/elsoc/> and the Harvard Dataverse repository <https://dataverse.harvard.edu/dataverse/elsoc>.

4.2 Variables

4.2.1 Dependent Variables: Justification of Political Violence

Two indices were constructed to measure attitudes toward political violence, capturing the justification of violence exercised by different types of actors in specific political contexts.

The Protest Violence Justification Index is composed of the average of two items measuring the degree to which respondents justify disruptive actions carried out by social actors in contexts of mobilization:

1. A group of striking workers blocking the street with barricades to demand compliance with their labor rights.
2. Students throwing stones at the Carabineros during a march for education.

Each item was measured on a 5-point Likert scale ranging from 1 = “Never justified” to 5 = “Always justified.” The index was calculated as the arithmetic mean of both items when the respondent answered at least one of them, yielding a continuous scale from 1 to 5 where higher values indicate greater justification of protest violence. The internal consistency of the index

is satisfactory, with a Cronbach's alpha of 0.82 in the analytical sample, indicating that the items capture a common latent construct.

The State Violence Justification Index is composed of the average of two items measuring the justification of violence exercised by the Carabineros de Chile (law enforcement) in public order control contexts:

1. The Carabineros using violence to repress a peaceful demonstration.
2. The Carabineros forcibly evicting people who have occupied buildings or universities.

Each item was measured on the same 5-point scale. The index was calculated as the mean of both items (requiring a response to at least one), also yielding a continuous scale from 1 to 5. The correlation between the two items is $r = 0.58$ ($p < 0.001$), indicating that they measure related but distinguishable aspects of attitudes toward police violence.

It is important to note that these two indices capture conceptually distinct forms of political violence: “bottom-up” violence exercised by mobilized social actors versus “top-down” violence exercised by state agents. As shown in the correlation matrix in Table 1, the two indices are negatively correlated ($r = -0.23$, $p < 0.001$), suggesting that justifying one type of violence tends to be associated with lower justification of the other, though the correlation is far from perfect. This empirical distinction validates the strategy of analyzing both types of violence separately.

4.2.2 Independent Variables: Education and Protest Participation

Respondent education was operationalized as a five-level categorical variable based on the highest level of education attained:

1. Complete secondary education or less
2. Incomplete tertiary non-university education (begun but not completed)
3. Complete tertiary non-university education (technical-professional degree or equivalent)
4. Incomplete university education (begun but not completed)
5. Complete university education (undergraduate or postgraduate degree)

This categorization is treated as an unordered factor, using “complete secondary education or less” as the reference category. The decision not to treat education as an ordinal variable reflects prior evidence suggesting that the effects of education on political attitudes are not necessarily linear, and that there are qualitative differences between technical and university education that go beyond years of schooling (Nie et al., 1996).

Protest participation was operationalized as a dichotomous variable indicating whether the respondent participated in demonstrations, marches, or protests during the previous 12 months (1 = Participated, 0 = Did not participate), based on the question: “During the last 12 months, have you participated in demonstrations or marches?” This variable captures retrospectively reported participation and is therefore subject to potential recall and social desirability biases, although prior evidence suggests that protest participation reports have reasonable validity when the reference period is relatively short (Walgrave et al., 2014).

4.2.3 Control Variables

All models include the following control variables to reduce confounding and improve the precision of the estimates:

- Age: In years at the time of each wave (continuous variable, centered at the mean).
- Gender: Dichotomous variable (1 = Woman, 0 = Man).
- Political ideology: Self-placement scale from 0 (far left) to 10 (far right), standardized to mean 0 and standard deviation 1.
- Time: Linear time trend (wave coded 0–6) to control for systematic changes in attitudes over the study period (2016–2023), including the impact of the 2019 social uprising.

4.3 Analytical Strategy

The longitudinal nature of the data implies that the same individuals are observed on multiple occasions over time. This data structure—in which Level 1 observations (measurements over time) are nested within Level 2 individuals—violates one of the fundamental assumptions of standard regression models: the independence of observations (Snijders et al., 2012). Responses from the same individual across waves are highly likely to be correlated, as they reflect stable characteristics of that person (personality, deep-seated values, prior formative experiences) that consistently shape their attitudes over time.

Ignoring this within-individual temporal dependence would yield biased estimates (standard errors that are too small), artificially inflating statistical significance and leading to invalid conclusions (Raudenbush et al., 2010; Gelman et al., 2021). Multilevel models (also referred to as hierarchical linear models, mixed-effects models, or variance component models) offer an explicit solution to this problem by simultaneously modeling both between-individual heterogeneity and within-individual dependence over time (Singer et al., 2003). As Hox et al. (Hox et al., 2018) note, these models allow the total variation to be decomposed into its structural components and the uncertainty of parameters to be correctly estimated in the presence of intra-group dependence.

The general model equations for Level 1 (observation over time) and Level 2 (individual) are as follows:

Level 1 (within-individual)

$$y_{tj} = \beta_{0j} + \beta_{1j}X_{tj} + \varepsilon_{tj}$$

where:

- y_{tj} : Justification of violence for individual j at time t .
- X_{tj} : Level 1 variables (time), such as participation in protests, age, etc.
- ε_{tj} : Residual error at the observation level.

Level 2 (between-individual)

$$\beta_{0j} = \gamma_{00} + \gamma_{01}Z_j + u_{0j}$$

$$\beta_{1j} = \gamma_{10}$$

where:

- γ_{00} : General intercept (overall mean).
- γ_{01} : Effect of Level 2 predictors (between individuals).

- γ_{10} : Fixed effect of Level 1 predictors (within individuals).
- Z_j : Predictors that do not vary over time (education, gender).
- u_{0j} : Random effect of the intercept for individual j .

Integrated Model

$$y_{tj} = \gamma_{00} + \gamma_{01}Z_j + \gamma_{10}X_{tj} + \gamma_{20}t + u_{0j} + \varepsilon_{tj}$$

where t represents time (measurement wave, coded as 0, 1, 2, ..., 6 for waves 2016-2023). This model decomposes the total variance into two components: variance between individuals (σ_u^2 , captured by the random intercepts) and variance within individuals over time (σ_ε^2 , residual error). The intraclass correlation coefficient (ICC) quantifies the proportion of total variance attributable to differences between individuals. The parameter γ_{20} captures the average linear temporal trend in the justification of violence.

4.3.1 Interaction Models

Interaction between Education and Participation

To test the central hypotheses of this study, which predict that the effects of education on the justification of violence are contingent upon participation in protests, we estimate models that include multiplicative interaction terms:

$$y_{tj} = \gamma_{00} + \sum_{k=1}^4 \gamma_{0k} \text{Education}_{kj} + \gamma_{10} \text{Participation}_{tj} + \sum_{k=1}^4 \gamma_{1k} (\text{Education}_{kj} \times \text{Participation}_{tj}) + \gamma_{20}t + \text{Controls} + u_{0j} + \varepsilon_{tj}$$

Given that education is a categorical variable with 5 levels (4 dummies), this model estimates 4 interaction coefficients ($\gamma_{11}, \gamma_{12}, \gamma_{13}, \gamma_{14}$), one for each educational level relative to the reference category.

The interpretation of interaction terms in multilevel models follows the standard logic: the interaction coefficient represents how much the effect of participation in protests on the justification of violence changes when moving from one educational category to another. Marginal effects plots (predicted values) are used to visualize these interactions in an intuitive manner.

4.3.2 Choice of Distribution and Link Function

The main dependent variable is the continuous violence justification index (scale 1–5). Although technically ordinal (discrete ordered levels), treating it as a continuous variable is justified on three methodological and substantive grounds. First, the indices average responses across multiple items (2 for protest violence, 2 for state violence), producing a more continuous and approximately normal distribution in the sample. Second, prior work in psychology and political science has shown that Likert scales with 5 or more points can be treated as continuous without substantial loss of information when multiple items are averaged (Norman, 2010; Lubke et al., 2004). Third, using continuous variables allows effects to be interpreted in terms of average changes on the justification scale, which is more intuitive and better captures subtle gradations in attitudes.

Table 1: Justification of violence by education level and protest participation

Education	Participation	N	Protest Violence		State Violence	
			Mean	SD	Mean	SD
High School or less	Yes	8139	1.71	0.83	1.87	1.02
High School or less	No	671	2.15	0.98	1.65	0.94
Technical College incomplete	Yes	448	1.70	0.79	1.93	1.03
Technical College incomplete	No	79	2.34	0.96	1.75	0.90
Technical College complete	Yes	2267	1.62	0.74	1.90	0.98
Technical College complete	No	446	2.15	0.98	1.59	0.82
University incomplete	Yes	739	1.75	0.78	1.89	0.97
University incomplete	No	292	2.29	1.03	1.52	0.77
University complete	Yes	2192	1.64	0.77	1.96	0.98
University complete	No	971	2.27	1.06	1.47	0.71

Note:

Compiled by the author based on ELSOC (2016–2023). Scale of 1–5, where 1 = Never justified and 5 = Always justified

The main models are therefore estimated using Generalized Linear Mixed Models (GLMM) with a Gaussian (normal) distribution and identity link function, incorporating random intercepts by individual to account for the hierarchical structure of the data. This specification is equivalent to a standard multilevel linear regression model and is estimated via restricted maximum likelihood (REML), which produces unbiased estimates of the variance components in finite samples (Raudenbush et al., 2010).

4.4 Software and Reproducibility

All statistical analyses were conducted in R (version 4.3.1) using the glmmTMB package (Brooks et al., 2017) for the estimation of generalized mixed models,ggeffects (Lüdtke, 2018) for the calculation of marginal effects and adjusted predictions, and ggplot2 (Wickham, 2016) for the visualization of results. Complete analysis scripts—including code for data cleaning, variable construction, model estimation, and figure generation—are available in an [OSF project](#), for transparency and replicability of the reported findings.

5 Results

This section presents the findings on the factors shaping the justification of violence in the context of social protest in Chile.

5.1 Descriptive Statistics

5.1.1 Justification of Violence by Education Level

The patterns observed in Table 1 reveal a notable trend. Among non-participants, higher education levels tend to be associated with lower justification of protest violence, with means declining from approximately 1.90 to 1.80. However, this trend reverses among those who do participate: participants with university education display the highest justification levels, reaching means around 2.29 and 2.27. This differentiated effect of education appears to be specific to protest violence, as, by contrast, the justification of state violence shows a much more stable pattern, with considerably smaller variation across educational levels.

5.1.2 Temporal Evolution: Justification of Violence and Protest Participation (2016–2023)

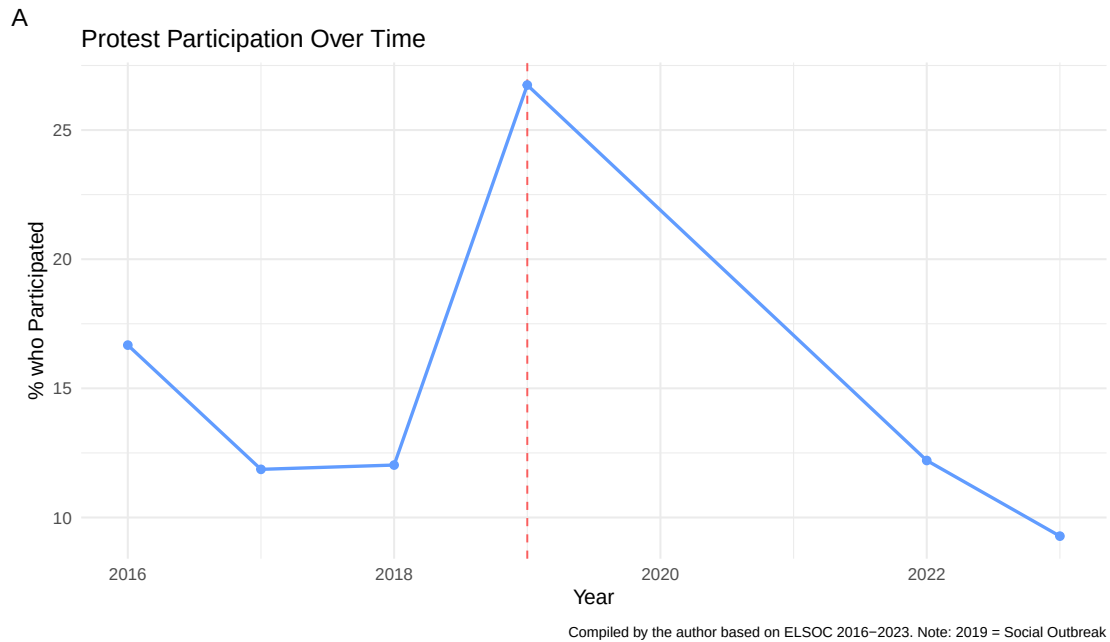


Figure 1: Evolution of protest participation 2016–2023

The temporal pattern analysis of Figure 1 shows a result consistent with the observed protest cycle context. In 2019 — the year of the social outbreak — the highest average protest participation rate is recorded, reflecting the intensity of mobilization. In the post-2019 period, although participation declines from its peak, it remains at slightly higher levels than those observed before 2019, until its fall between 2021 and 2023, where it reaches its lowest average across the entire reported cycle.

Divergence between Justification of Violence in Protests and State Violence

Crossed lines post-2019 reveal inversion of moral hierarchies: protests more legitimized than repression

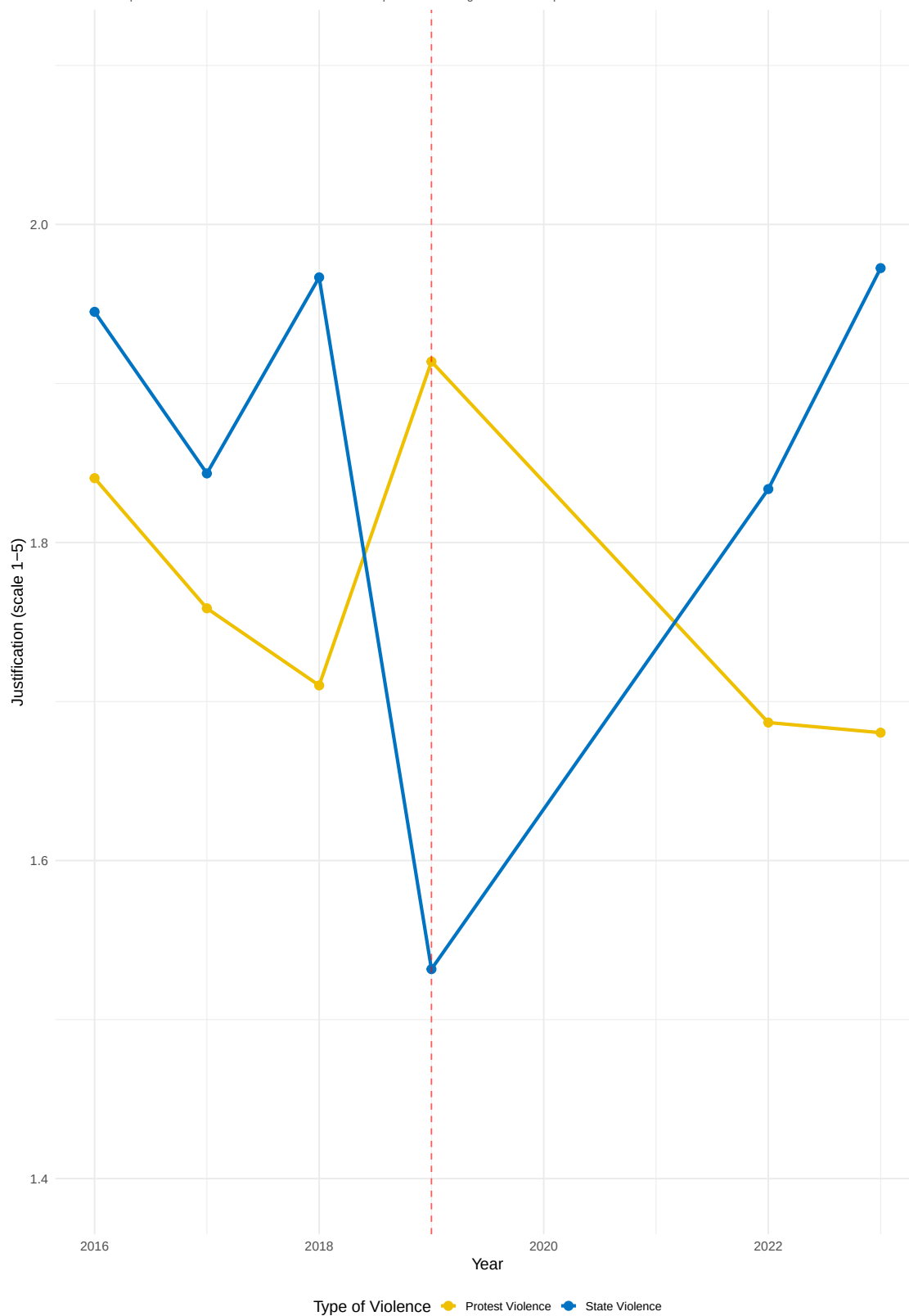


Figure 2: Sociodemographic composition of protest participants by year: structural changes post-outbreak. Compiled by the author based on ELSOC.

Figure 2 reveals the radical transformation of moral hierarchies around political violence. Pre-2019 (2016–2018), both forms of violence were evaluated with similar criteria (gap of ± 0.05 points), reflecting liberal-democratic consensus where violence “from below” and “from above” were equally rejected (~ 1.6 – 1.7). In 2019, the gap reverses dramatically: state violence was justified ~ 0.15 points more than protest violence, reflecting the initial shock and polarization in response to perceived “chaos.” However, in the post-2019 period (2022–2023), the lines cross definitively: protest violence is justified ~ 0.10 – 0.15 points more than state violence, and this gap stabilizes. This is the critical finding: the outbreak generated a permanent reconfiguration of collective moral hierarchies, in which disruptive tactical violence is normalized more than state repression. This pattern evidences the bidirectional reconfiguration: the emerging dual moral framework post-outbreak redefines the boundaries of legitimate political action depending on who exercises violence and in what context.

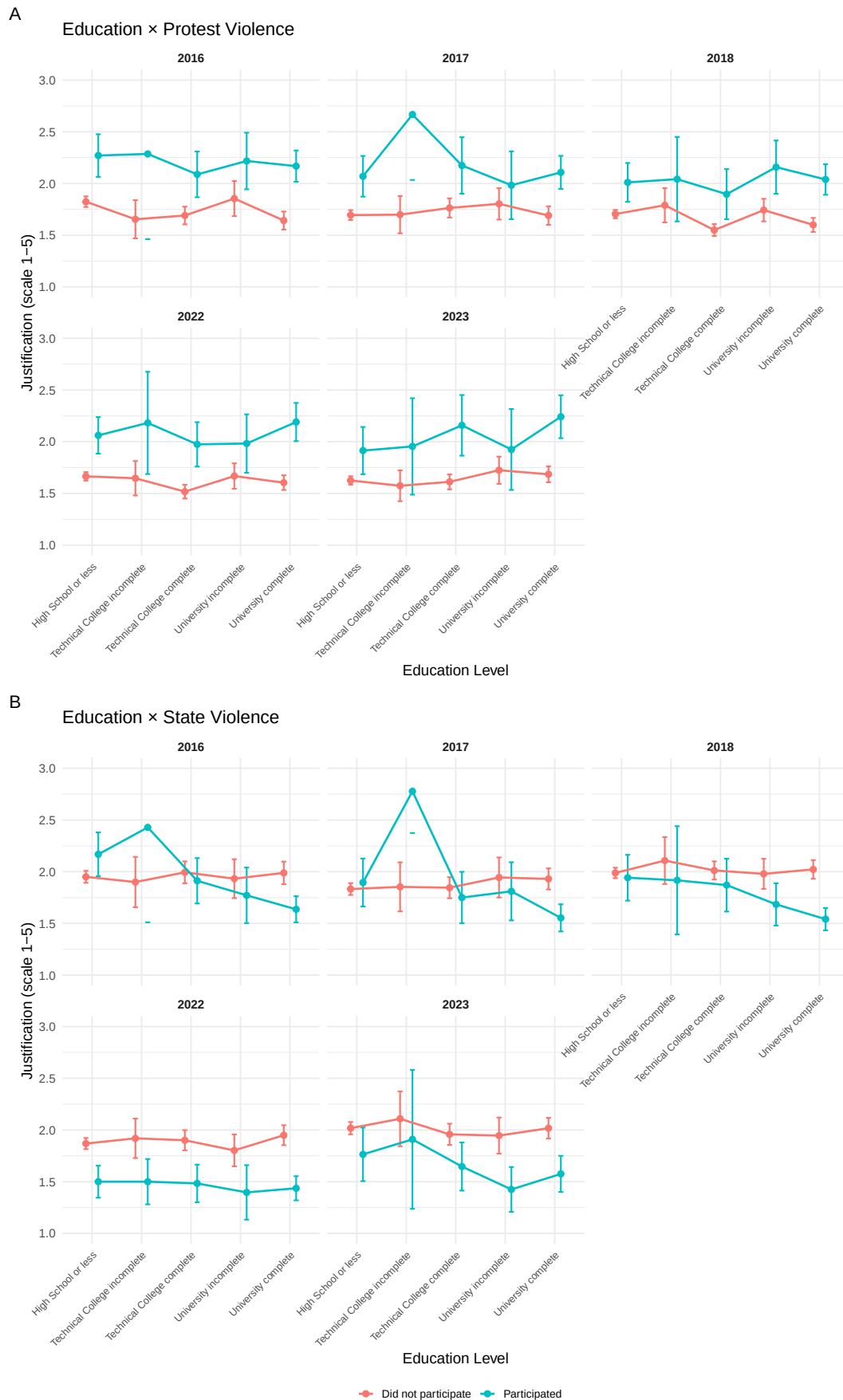


Figure 3: Comparison of violence justification before (2016–2018) and after (2022–2023) the social outbreak by education and participation

Figure 3 presents the temporal evolution of violence justification by education, comparing year-by-year levels before (2016–2018) and after (2022–2023) the October 2019 social outbreak, revealing critical patterns about the persistence of the political shock effects.

Among non-participants, relative stability is observed in the pre-2019 period, with justification levels oscillating between ~ 1.5 – 1.8 for higher education and ~ 1.7 – 2.0 for lower education. In the post-2019 period (2022–2023), all educational groups show a moderate increase: university completers rise from ~ 1.5 to ~ 1.8 , while basic/secondary education remains in the ~ 1.8 – 2.0 range. Among participants, the pattern shows sustained convergence: from 2016 to 2023, all educational levels maintain moderately high justifications (~ 2.0 – 2.3) with overlapping confidence intervals, indicating that the experience of mobilization equalizes attitudes regardless of cultural capital. Crucially, the gaps between participants and non-participants remain consistent year after year in the post-2019 period, showing that the participation effect persists stably years after the outbreak.

The contrast reveals that among non-participants, a dramatic jump is observed between 2018 and 2022 across all educational groups: university completers go from ~ 1.3 – 1.5 (2016–2018) to ~ 1.9 – 2.0 (2022–2023). This is the most critical finding: the outbreak did not generate permanent rejection of state repression, but rather normalized its justification even among privileged sectors, and this normalization remains stable in 2022 and 2023. Among participants, the pattern reverses completely: all educational groups show sustained declines post-2019, going from ~ 1.8 – 2.0 (2016–2018) to ~ 1.5 – 1.7 (2022–2023), with university completers maintaining consistently low levels (~ 1.4). The year-by-year trajectory confirms that direct experience of repression during the outbreak generated a permanent “experiential delegitimization” effect among demonstrators, consistently observable in both post-outbreak years.

The data reveal that the 2019 outbreak was not a temporary shock but a permanent inflection point that structurally reconfigured dual moral frameworks. The year-by-year visualization demonstrates that the divergence between participants and non-participants is not transient fluctuation but stable polarization: while participants develop lasting rejection of state violence yet maintain justification of protest violence, non-participants experience divergent patterns depending on cultural capital. These frameworks remain stable 3–4 years after the critical event, crystallizing the hypothesis of permanent bidirectional polarization.

5.2 Exploratory Statistics

Table 2 presents two main models guiding the analysis of this study. Model 1 examines the effects of education and protest participation on protest violence justification, and Model 2 evaluates those same effects on state violence justification. Both models include random effects by individual and controls for age, gender, ideology, and year.

The results reveal differentiated patterns depending on the type of violence. For protest violence, participation shows a consistent positive effect, increasing justification by approximately 0.36 points. By contrast, for state violence, the direct effect of participation is negative and highly significant, showing that protest participants do not more readily justify police repression.

The controls operate consistently: right-wing ideology reduces justification of protest violence (~ -0.03 per point on the scale) but increases it for state violence ($\sim +0.05$). Women reject state violence more (~ -0.13) with no differences on protest violence. The year 2019 shows the most dramatic contrast: a moderate increase in protest violence justification but a sharp fall in state violence justification (~ -0.40), reflecting the impact of the social outbreak on attitudes toward police repression.

Table 2: Main models of violence justification

	Model 1	Model 2
Education (Ref. = High School or less)		
Technical College incomplete	-.096 (.050)	.106 (.059)
Technical College complete	-.121*** (.025)	.030 (.029)
University incomplete	-.054 (.038)	.011 (.044)
University complete	-.105*** (.025)	.023 (.029)
Controls		
Protest participation	.357*** (.020)	-.133*** (.023)
Age	-.009*** (.001)	.004*** (.001)
Woman	-.028 (.018)	-.133*** (.021)
Ideology	-.028*** (.003)	.055*** (.004)
Year (Ref. = 2016)		
2017	-.043 (.023)	-.107*** (.026)
2018	-.108*** (.021)	.011 (.024)
2019	.049* (.021)	-.401*** (.024)
2022	-.101*** (.023)	-.128*** (.026)
2023	-.099*** (.023)	-.002 (.026)
Num. obs.	3383	3383
Num. groups: idencuesta	.134	.197
AIC	36296.693	40375.594
BIC	15110	15103

*** p < 0.001; ** p < 0.01; * p < 0.05

Table 3: Statistical models

Table 4: Interaction models for violence justification

	Model 1b	Model 2b
Education (Ref. = High School or less)		
Technical College incomplete	-.133* (.053)	.095 (.062)
Technical College complete	-.134*** (.026)	.032 (.031)
University incomplete	-.067 (.041)	.038 (.048)
University complete	-.126*** (.026)	.059 (.031)
Protest participation	.265*** (.035)	-.042 (.040)
Education Interactions		
Technical College incomplete $\times$ Protest	.281* (.111)	.021 (.127)
Technical College complete $\times$ Protest	.129* (.056)	-.063 (.064)
University incomplete $\times$ Protest	.109 (.068)	-.159* (.078)
University complete $\times$ Protest	.133** (.048)	-.183*** (.055)
Controls		
Age	-.009*** (.001)	.004*** (.001)
Woman	-.028 (.018)	-.133*** (.021)
Ideology	-.027*** (.003)	.055*** (.004)
Year (Ref. = 2016)		
2017	-.043 (.023)	-.107*** (.026)
2018	-.108*** (.021)	.011 (.024)
2019	.049* (.021)	-.402*** (.024)
2022	-.101*** (.023)	-.129*** (.026)
2023	-.098*** (.023)	-.005 (.026)
Num. obs.	3383	3383
Num. groups: idencuesta	.133	.197
AIC	36291.927	40370.252
BIC	15110	15103

*** p < 0.001; ** p < 0.01; * p < 0.05

Table 5: Statistical models

The models in Table 4 reveal that the effect of participation varies significantly by education level. For protest violence, the interactions between education and participation are positive, indicating that the legitimizing effect of participating is stronger among more educated individuals. A person with complete university education who participates increases their justification by 0.40 points versus 0.27 for secondary education, evidencing the “educational paradox”: education reduces justification among non-participants (-0.13***) but amplifies the effect of participation.

For state violence, the pattern reverses completely. The interactions are negative (university incomplete: -0.16; *university complete*: -0.18**), indicating that only the combination of university education + participation generates significant rejection of repression (total effect: -0.22 points). Education alone (+0.06, ns) or participation alone (-0.04, ns) are insufficient; it is their interaction that activates the differentiated rejection.

The bidirectional reconfiguration (legitimizing protests, rejecting repression) is clearest among university-educated participants, confirming that the dual moral framework requires the critical combination of higher education and direct experience of mobilization.

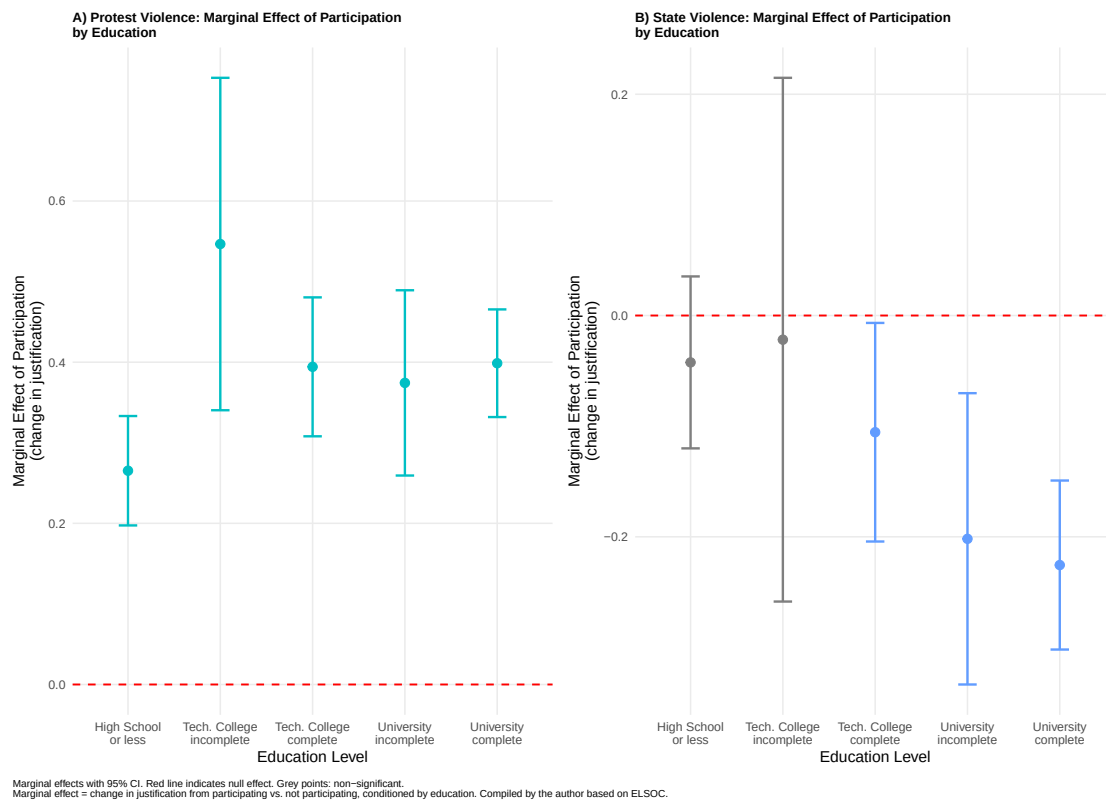


Figure 4: Marginal effects of protest participation on violence justification by education

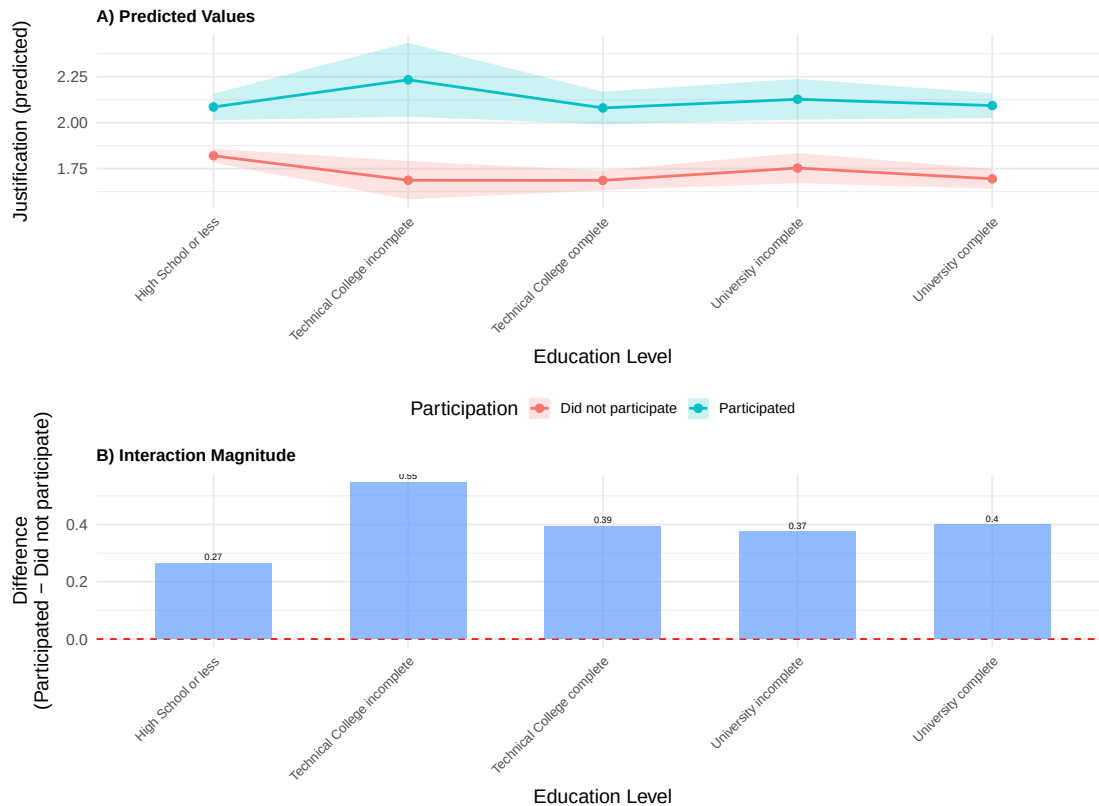
Figure 4 visualizes the marginal effects of participation by education, allowing direct evaluation of the hypotheses about differential effects.

In Panel A, all education levels show significant positive effects on protest violence justification, with magnitudes ranging from +0.27 (secondary education) to +0.55 (technical college incomplete). Crucially, the legitimizing effect is stronger among technical college (+0.55) and university (+0.40) education than among secondary education (+0.27), showing that education

amplifies (rather than reduces) the effect of participation, inverting the civilizing effect among those who mobilize.

Panel B shows that only university-educated individuals have significant negative effects on state violence justification (-0.20 to -0.22), while secondary and technical education show null effects (grey points on the red line). University-educated participants thus tend to legitimize protest violence (+0.40) while simultaneously rejecting state violence (-0.22).

5.2.1 The Education “Paradox Effect”



Panel A shows effects by group; Panel B quantifies the interaction. Compiled by the author based on ELSOC 2016–2023.

Figure 5: Interaction analysis: Education × Protest Participation

Figure 5 reveals the “educational paradox” predicted in hypotheses H1a and H1b. Panel A shows that education reduces violence justification among external observers (descending red line from ~1.95 at secondary education to ~1.75 at university complete), consistent with the civilizing effect. However, education inverts its effect among participants (ascending blue line from ~2.20 at secondary education to ~2.35 at technical college incomplete, then ~2.30 at university complete), showing that more educated individuals construct ideological frameworks that legitimize tactical violence in mobilization contexts. Panel B quantifies this inversion: the interaction magnitude is highest at technical college incomplete (+0.55) and university incomplete (+0.45), confirming that the “paradox” is most pronounced at mid-to-upper education levels where cultural capital and mobilization experience converge.

5.3 Bidirectional Reconfiguration of Violence Justification (Protest Violence vs. State Violence)

The interaction models (Model 1b and Model 2b) reveal the bidirectional dynamic predicted in H2a–H2b (university-educated). H2a is fully confirmed: for protest violence, university education reduces justification among non-participants (-0.13^{**}), but participation generates an increase ($+0.27^{**}$) magnified by positive interactions with higher education (technical college incomplete: $+0.28^{*}$; university complete: $+0.13^{*}$), nullifying the civilizing effect. H2b is fully confirmed: for state violence, the pattern reverses completely: education alone ($+0.06$, ns) and participation alone (-0.04 , ns) generate no effects, but their negative interactions produce significant rejection among university-educated individuals (incomplete: -0.16^{*} ; complete: -0.18^{**}), showing that only the combination of university education + participation activates the critical framework toward repression.

The controls confirm the impact of the 2019 outbreak: moderate increase in protest justification ($+0.05^{*}$) but collapse in state violence (-0.40^{***}), radically transforming attitudes toward police repression. Ideology operates in opposite directions: right-wing ideology reduces justification of protests (-0.03^{**}) but increases that of state violence ($+0.05^{**}$). Women reject state violence more (-0.13^{**}) with no differences in protests. This pattern confirms that the bidirectional reconfiguration predicted in H2a–H2b (legitimizing protests, rejecting repression) is complete only among those who combine university education with direct experience of mobilization, crystallizing the sophisticated dual moral framework.

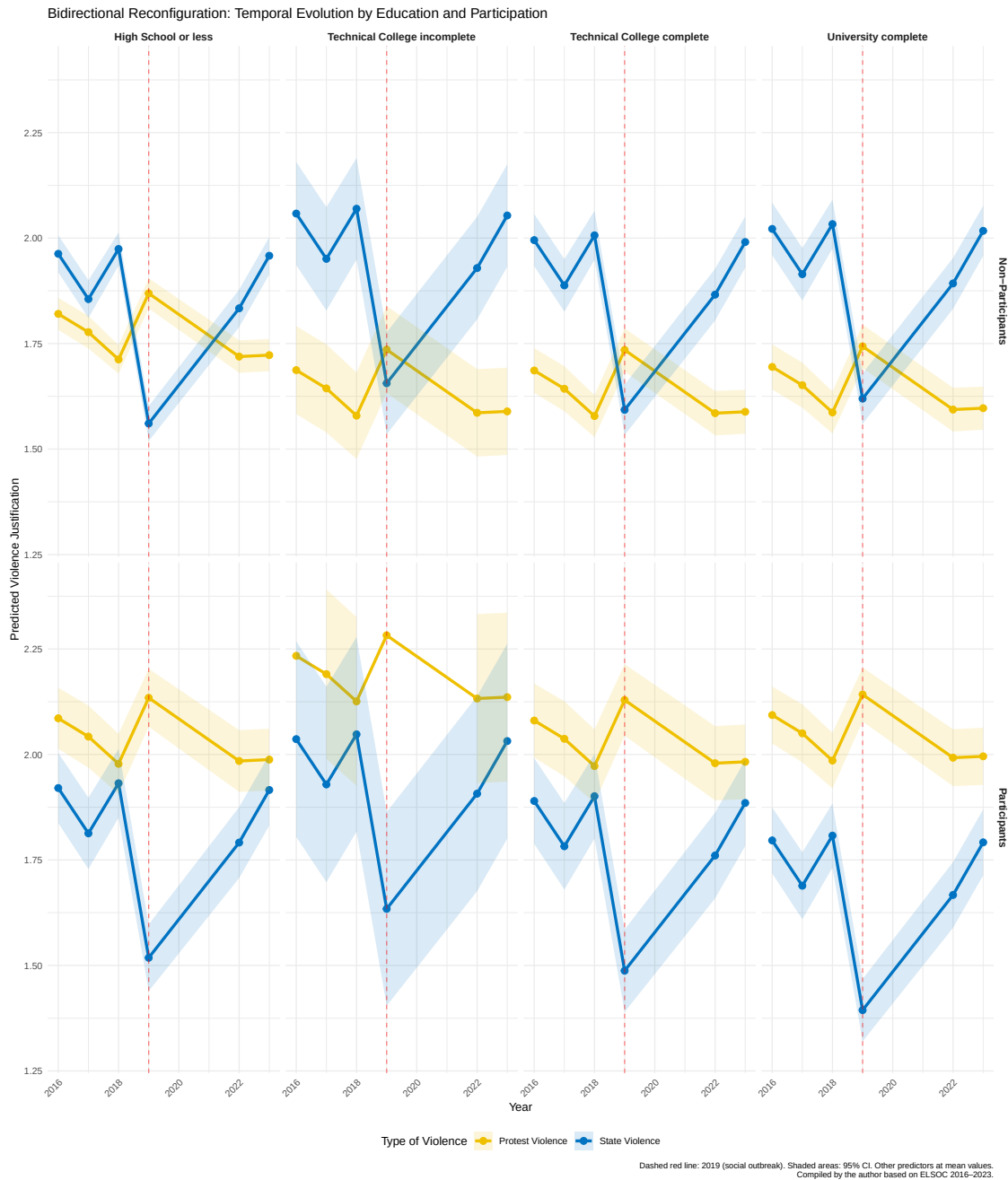


Figure 6: Temporal evolution of bidirectional reconfiguration by education and participation (2016–2023)

Figure 6 visualizes the temporal dynamics of the bidirectional reconfiguration predicted in H2a–H2b, showing that the dual moral framework requires both education and participation. Among non-participants (top row), the yellow (protests) and blue (state) lines remain parallel and close across all education levels, confirming that without participation there is no bidirectional reconfiguration despite the university civilizing effect (H1a operates but H2b is not activated).

Among participants (bottom row), three patterns emerge for evaluating H2a–H2b. Secondary/technical education shows partial rejection of H2a–H2b: elevated protest violence justification (+0.27 to +0.55, confirming H2a) but without significant rejection of state violence (rejecting H2b), evidencing tactical legitimization without a critical framework toward repression.

Technical college incomplete reaches the highest radicalization on protests (>2.2 in 2019, $+0.28^*$) but maintains parallel lines, confirming that technical capital without university socialization does not generate the normative differentiation predicted in H2b.

The complete dual framework predicted in H2a–H2b is confirmed exclusively among university-educated participants: (1) a sustained elevated yellow line (~ 2.0 – 2.1), nullifying the civilizing effect and confirming H2a; (2) a dramatic collapse of the blue line in 2019 from ~ 1.75 to ~ 1.35 , the lowest level in the figure (-0.40^{***}), confirming H2b; (3) maximum post-2019 divergence with a gap of ~ 0.7 points, evidencing the permanence of the dual framework. This pattern confirms that H2a and H2b are jointly satisfied only under the critical combination of university education + mobilization experience, crystallizing the sophisticated dual normative framework.

6 Discussion and Conclusions

As outlined at the outset, this article departs from a fundamental question raised by the 2019 social uprising: why and how did sectors traditionally associated with political moderation and institutional attachment come to legitimize disruptive forms of political violence? The findings challenge classical linear conceptions of education's role, proposing instead a model in which protest participation operates as an activation and transformation mechanism that inverts the expected effects of cultural capital.

The classical literature in political sociology has long defended the “civilizing effect” hypothesis (Lipset, 1959; Nie et al., 1996), premised on the idea that education fosters tolerance and rejection of violence as a mode of political action. The findings of this study confirm this relationship, but with a critical qualification: it holds only for those who observe politics from a distance.

Among non-participants, higher education is indeed associated with lower justification of violence. However, once the experience of protest participation is introduced, this relationship reverses dramatically — confirming Hypothesis 1 (H1). University-educated participants not only justify violence more than their non-mobilized peers; they reach the highest justification levels in the entire sample, surpassing even those with the lowest levels of formal education.

This finding suggests that education does not operate solely as a universal moral brake, but also as a conditional cognitive amplifier. The cognitive sophistication conferred by higher education does not determine what to think — that is, it does not produce automatic rejection of violence — but rather provides complex tools for processing political experience. When that experience involves confrontation with the state and participation in a high-intensity cycle such as that of 2019, educated individuals draw on their cultural capital not to moderate their views, but to construct sophisticated interpretive frameworks (Snow et al., 1988) that legitimize tactical violence as a necessary resource of resistance or self-defense.

This socialization-based interpretation must, however, be qualified by a critical consideration regarding causal direction. The results do not permit ruling out an alternative selection-based explanation: it is possible that university students who choose to participate in protests already hold more radical or institutionally critical prior dispositions, and that these pre-existing differences — rather than participation per se — account for part of the higher justification of violence observed. In other words, university education may be “selecting” and concentrating individuals with specific political predispositions that are

subsequently reflected in their participation and attitudes, rather than directly transforming them through the experience of mobilization.

The study design — which includes individual-level controls through random effects and political ideology variables — partially addresses this concern by isolating within-person variation over time. Nevertheless, the possibility of unobserved heterogeneity (e.g., latent political radicalism, activist social networks, or prior exposure to police violence) that may simultaneously correlate with university education, the decision to participate, and the justification of violence cannot be fully eliminated without a more robust experimental or quasi-experimental design. Future research could address this question through longitudinal designs that measure attitudes before and after first-time protest participation, or through matching techniques that compare participants and non-participants with identical observable characteristics.

The cognitive amplification interpretation also requires an analytical distinction that the current data do not fully allow to explore: the difference between those actively embedded in university contexts (enrolled students) and those who hold university credentials but have left campus (graduates or dropouts). The literature on university socialization ([Gurin et al., 2002](#)) suggests that the formative effect of higher education resides not solely in the accumulation of abstract cultural capital, but in prolonged immersion in spaces of political deliberation, ideological diversity, and student activism that characterize campus life.

If cognitive amplification operates primarily through the political socialization that occurs within universities — where critical debates, student organizations, and activist networks converge — then enrolled students would be expected to display qualitatively different patterns from university graduates who have transitioned to the labor market and left those spaces of political effervescence. The former not only possess the cognitive tools to process protest, but are immersed in social ecologies where participation in mobilizations is normalized, discussed, and collectively organized. The latter, despite having acquired university cultural capital, no longer participate in those spaces of continuous socialization and may exhibit attenuated effects. This distinction is especially relevant in the Chilean context, where the university student movement has historically been a central actor in social protest (2006, 2011, 2019). Future research should disaggregate the “university-educated” category into enrolled students, recent graduates, and those who graduated more than five years ago, in order to assess whether the amplification effect is stronger among those who remain on campus, declines over time following graduation, or persists as a permanent effect of educational credentialism independent of institutional context.

Political participation thus appears to disrupt the civilizing effect of education. For mobilized university students, violence seems to undergo a process of re-moralization — acquiring a political meaning that renders it compatible with their abstract democratic values and lending it a situated, tactical justification. This transformation, however, may be conditioned by experiential socialization processes (the causal effect of participating), self-selection processes (who chooses to participate), and ongoing institutional socialization (who remains in active university contexts) — mechanisms this study cannot fully disentangle, but which represent critical avenues for future research. Perhaps the most substantive finding for understanding contemporary Chilean conflict is the confirmation of a bidirectional reconfiguration of attitudes. Contrary to the view that the uprising generated generalized anomie or indiscriminate validation of all forms of violence, the data reveal the emergence of a highly sophisticated dual moral framework — particularly among educated participants.

Participation simultaneously produces an increase in the justification of protest violence and a sharp decrease in the justification of state violence. This pattern dissolves the positive correlation that had previously existed between both types of violence

during periods of normality, in which those who rejected violence tended to reject all forms of it. The study demonstrates that protesters — and in particular those who combine university education with active participation — operated under a logic of selective delegitimation during the most intense moment of the conflict. In 2019, while validating disruptive violence as a legitimate political tool (presumably in response to a perceived institutional vacuum), they simultaneously withdrew legitimacy from the use of force by the national police (Carabineros). This is consistent with procedural justice theory (Tyler, 2006): the direct experience of repression perceived as unjust or disproportionate does not produce submission, but rather temporarily erodes the state's moral authority, rendering its violence illegitimate and one's own violence defensive.

The temporal dynamics of this effect, however, reveal a more complex picture than a permanent fracture. Post-2019 data (2022–2023 waves) show a progressive convergence between both types of violence among participants: while justification of protest violence declines from its 2019 peak, justification of state violence gradually increases, narrowing the bidirectional gap that had reached its maximum during the uprising. This post-crisis convergence suggests that the dual moral framework did not consolidate as a new permanent political subjectivity, but rather responded reactively to the specific political context of 2019, partially receding as the intensity of the conflict subsided.

This temporal pattern is consistent with models of context effects, in which acute political crises generate temporary attitudinal realignments that attenuate once the moment of collective effervescence declines (Snow & Moss, 2014). The narrowing of the post-2019 bidirectional gap among educated participants suggests that, while the experience of mobilization produced a profound transformation in that moment, it did not crystallize into a permanent rupture of the social contract. Instead, both participants and non-participants have undergone a process of post-uprising normalization, though from differentiated starting points: participants reduce their legitimization of protest violence while increasing their tolerance of state repression, converging toward levels closer to those observed before 2019.

It is important to note, however, that convergence does not imply a full return to the pre-uprising status quo. The 2022–2023 trajectories still differ from the pre-2019 period, and the experience of the uprising has left an identitarian mark on participants, particularly among the more educated sectors. The study shows that, during the critical moment of 2019, political violence was legitimized primarily by the most educated and socially integrated participants — challenging conventional readings that attribute political violence to marginality. Yet this legitimization was predominantly conjunctural, responding to the specific context of police repression and institutional crisis, rather than consolidating as a permanent cultural transformation of the moral rules governing the use of force.

7 References

- ALMOND, G. A. and VERBA, S., *The Civic Culture*, Princeton University Press, accessed November 24, 2025, from <https://www.jstor.org/stable/j.ctt183pnr2>, 1963.
- Amnistía, I., Ojos Sobre Chile: Violencia Policial y Responsabilidad de Mando Durante El Estallido Social, AMR 22/3133/2020, 2020.
- Araujo, K., *The Circuit of Detachment in Chile: Understanding the Fate of a Neoliberal Laboratory*, Cambridge, United Kingdom ; New York, NY: Cambridge University Press, 2022.
- Bellei, C., Cabalin, C. and Orellana, V., The 2011 Chilean Student Movement Against Neoliberal Educational Poli-

- cies, *Studies in Higher Education*, vol. **39**, no. 3, pp. 426–40, accessed November 24, 2025, March 2014. DOI: [10.1080/03075079.2014.896179](https://doi.org/10.1080/03075079.2014.896179)
- Benford, R. D. and Snow, D. A., Framing Processes and Social Movements: An Overview and Assessment, *Annual Review of Sociology*, vol. **26**, pp. 611–39, accessed November 24, 2025, from <https://www.jstor.org/stable/223459>, 2000.
- Berinsky, A. J. and Lenz, G. S., Education and Political Participation: Exploring the Causal Link, *Political Behavior*, vol. **33**, no. 3, pp. 357–73, accessed November 24, 2025, from <https://www.jstor.org/stable/41488848>, 2011.
- Bonner, M. D. and Dammert, L., Constructing Police Legitimacy During Protests: Frames and Consequences for Human Rights, *Policing and Society*, vol. **32**, no. 5, pp. 629–45, accessed November 24, 2025, May 2022. DOI: [10.1080/10439463.2021.1957887](https://doi.org/10.1080/10439463.2021.1957887)
- Brady, H. E., Verba, S. and Schlozman, K. L., Beyond SES: A Resource Model of Political Participation, *American Political Science Review*, vol. **89**, no. 2, pp. 271–94, accessed April 8, 2026, June 1995. DOI: [10.2307/2082425](https://doi.org/10.2307/2082425)
- Brooks, M., E., Kristensen, K., Benthem, van, J., Magnusson, A., Berg, C., W., Nielsen, A., Skaug, H., J., Mächler, M. and Bolker, B., M., glmmTMB Balances Speed and Flexibility Among Packages for Zero-inflated Generalized Linear Mixed Modeling, *The R Journal*, vol. **9**, no. 2, p. 378, accessed November 24, 2025, 2017. DOI: [10.32614/RJ-2017-066](https://doi.org/10.32614/RJ-2017-066)
- Canales Cerón, M., *La pregunta de Octubre: fundación, apogeo y crisis del Chile neoliberal*, Santiago de Chile: LOM ediciones, 2022.
- Carrasco Paillamilla, S. F. and Disi Pavlic, R., Predictors of Perceptions of Human Rights Violations During the Chilean Social Outburst of 2019, *Frontiers in Psychology*, vol. **14**, p. 1133428, accessed November 24, 2025, April 2023. DOI: [10.3389/fpsyg.2023.1133428](https://doi.org/10.3389/fpsyg.2023.1133428)
- Converse, P., Change in the American Electorate, *The Human Meaning of Social Change/Russell Sage Foundation*, 1972.
- Cox, L., González, R. and Le Foulon, C., The 2019 Chilean Social Upheaval: A Descriptive Approach, *Journal of Politics in Latin America*, vol. **16**, no. 1, pp. 68–89, accessed November 24, 2025, April 2024. DOI: [10.1177/1866802X231203747](https://doi.org/10.1177/1866802X231203747)
- Crossley, N., From Reproduction to Transformation: Social Movement Fields and the Radical Habitus, *Theory, Culture & Society*, vol. **20**, no. 6, pp. 43–68, accessed November 24, 2025, December 2003. DOI: [10.1177/0263276403206003](https://doi.org/10.1177/0263276403206003)
- Dahlum, S., Students in the Streets: Education and Nonviolent Protest, *Comparative Political Studies*, vol. **52**, no. 2, pp. 277–309, accessed April 8, 2026, February 2019. DOI: [10.1177/0010414018758761](https://doi.org/10.1177/0010414018758761)
- Davenport, C., Multi-Dimensional Threat Perception and State Repression: An Inquiry into Why States Apply Negative Sanctions, *American Journal of Political Science*, vol. **39**, no. 3, p. 683, accessed November 24, 2025, from <https://www.jstor.org/stable/2111650>, August 1995. DOI: [10.2307/2111650](https://doi.org/10.2307/2111650)
- Davenport, C., Soule, S. A. and Armstrong, D. A., Protesting While Black? The Differential Policing of American Activism, 1960 to 1990, *American Sociological Review*, vol. **76**, no. 1, pp. 152–78, accessed November 24, 2025, from <https://www.jstor.org/stable/25782184>, 2011.
- della Porta, D. Ed., *Global Diffusion of Protest: Riding the Protest Wave in the Neoliberal Crisis*, Amsterdam University Press, accessed November 24, 2025, from <https://www.jstor.org/stable/j.ctt1zkjxq0>, 2017.
- Della Porta, D. and Fillieule, O., [Policing Social Protest](#), in *The Blackwell Companion to Social Movements*, D. A. Snow S. A. Soule and H. Kriesi, Eds., Oxford, UK: Blackwell Publishing Ltd, accessed November 24, 2025, pp. 217–41, 2007.
- Dewey, J. and Tampio, N., *Democracy and Education*, New York: Columbia University Press, 2024.
- Disi Pavlic, R., The Nearness of Youth: Spatial and Temporal Effects of Protests on Political Attitudes in Chile, *Latin American*

- Politics and Society*, vol. **63**, no. 1, pp. 72–94, accessed April 8, 2026, February 2021. DOI: [10.1017/lap.2020.33](https://doi.org/10.1017/lap.2020.33)
- Disi Pavlic, R., Medel, R. M., Bargsted, M. and Somma, N. M., Justification of Violence, Ideological Preferences, and Exposure to Protests: Causal Evidence from the 2019 Chilean Social Unrest, *Social Forces*, p. soaf102, accessed November 24, 2025, July 2025. DOI: [10.1093/sf/soaf102](https://doi.org/10.1093/sf/soaf102)
- Donoso, S., Pavlic, R. D. and Cumsille R., B., Social Policy Expansion from Below? The Case of Chile's Student Movement and Free Tuition Higher Education, *World Development*, vol. **167**, p. 106213, July 2023. DOI: [10.1016/j.worlddev.2023.106213](https://doi.org/10.1016/j.worlddev.2023.106213)
- Dyrstad, K. and Hillesund, S., Explaining Support for Political Violence, *The Journal of Conflict Resolution*, vol. **64**, no. 9, pp. 1724–53, accessed November 24, 2025, from <https://www.jstor.org/stable/48631689>, 2020.
- Ellner, S., The Contrasting Variants of the Populism of Hugo Chávez and Alberto Fujimori, *Journal of Latin American Studies*, vol. **35**, no. 1, pp. 139–62, accessed June 24, 2026, February 2003. DOI: [10.1017/S0022216X02006685](https://doi.org/10.1017/S0022216X02006685)
- Fair, C. C. and Shepherd, B., Who Supports Terrorism? Evidence from Fourteen Muslim Countries, *Studies in Conflict & Terrorism*, vol. **29**, no. 1, pp. 51–74, accessed November 24, 2025, January 2006. DOI: [10.1080/10576100500351318](https://doi.org/10.1080/10576100500351318)
- Francisco, R. A., The Relationship Between Coercion and Protest: An Empirical Evaluation in Three Coercive States, *The Journal of Conflict Resolution*, vol. **39**, no. 2, pp. 263–82, accessed November 24, 2025, from <https://www.jstor.org/stable/174413>, 1995.
- Garcés, M., October 2019: Social Uprising in Neoliberal Chile, *Journal of Latin American Cultural Studies*, vol. **28**, no. 3, pp. 483–91, accessed November 24, 2025, July 2019. DOI: [10.1080/13569325.2019.1696289](https://doi.org/10.1080/13569325.2019.1696289)
- Gelman, A. and Hill, J., *Data Analysis Using Regression and Multilevel/Hierarchical Models*, Cambridge: Cambridge Univ. Press, 2021.
- Gerber, Monica M. and Jackson, J., Justifying Violence: Legitimacy, Ideology and Public Support for Police Use of Force, *Psychology, Crime & Law*, vol. **23**, no. 1, pp. 79–95, accessed November 24, 2025, January 2017. DOI: [10.1080/1068316X.2016.1220556](https://doi.org/10.1080/1068316X.2016.1220556)
- Gerber, Mónica M., Figueiredo, A., Sáez, L. and Orchard, M., Legitimidad, Justicia y Justificación de La Violencia Intergrupal Entre Carabineros y Manifestantes En Chile, *Psyke (Santiago)*, vol. **32**, pp. 0–0, 2023.
- Giugni, M., *Social Protest and Policy Change: Ecology, Antinuclear, and Peace Movements in Comparative Perspective*, Lanham: Rowman & Littlefield, 2004.
- Gonzalez, R. and Morán, C. L. F., The 2019–2020 Chilean Protests: A First Look at Their Causes and Participants, *International Journal of Sociology*, vol. **50**, no. 3, pp. 227–35, accessed November 24, 2025, May 2020. DOI: [10.1080/00207659.2020.1752499](https://doi.org/10.1080/00207659.2020.1752499)
- Górecki, M. A., Electoral Context, Habit-Formation and Voter Turnout: A New Analysis, *Electoral Studies*, vol. **32**, no. 1, pp. 140–52, accessed June 24, 2026, March 2013. DOI: [10.1016/j.electstud.2012.09.005](https://doi.org/10.1016/j.electstud.2012.09.005)
- Gurin, P., Dey, E., Hurtado, S. and Gurin, G., Diversity and Higher Education: Theory and Impact on Educational Outcomes, *Harvard Educational Review*, vol. **72**, no. 3, pp. 330–67, accessed March 23, 2026, September 2002. DOI: [10.17763/haer.72.3.01151786u134n051](https://doi.org/10.17763/haer.72.3.01151786u134n051)
- Hox, J. J., Moerbeek, M. and Schoot, R. van de, *Multilevel Analysis: Techniques and Applications*, New York, NY: Routledge, Taylor & Francis Group, 2018.
- Human, R. W., Chile: Llamado Urgente a Una Reforma Policial Tras Las Protestas., November 2019.

- Jämte, J., Lundstedt, M. and Wennerhag, M., When Do Radical Flanks Use Violence? Conditions for Violent Protest in Radical Left-Libertarian Activism in Sweden, 1997–2016, *Terrorism and Political Violence*, vol. **36**, no. 5, pp. 679–98, accessed June 24, 2026, July 2024. DOI: [10.1080/09546553.2023.2189966](https://doi.org/10.1080/09546553.2023.2189966)
- Joignant, A. and Somma, N. M. Eds., *Social Protest and Conflict in Radical Neoliberalism: Chile, 2008–2020*, Cham: Springer Nature Switzerland, accessed November 24, 2025, 2024.
- Joignant, A., Somma, N., Garretón, M., and Capos, T., Informe Anual Observatorio de Conflictos 2020, Centro de Estudios de Conflicto y Cohesión Social (COES), 2020.
- Kam, C. D. and Palmer, C. L., Reconsidering the Effects of Education on Political Participation, *The Journal of Politics*, vol. **70**, no. 3, pp. 612–31, accessed November 24, 2025, July 2008. DOI: [10.1017/S0022381608080651](https://doi.org/10.1017/S0022381608080651)
- Koopmans, R., The Dynamics of Protest Waves: West Germany, 1965 to 1989, *American Sociological Review*, vol. **58**, no. 5, p. 637, accessed November 24, 2025, from <https://www.jstor.org/stable/2096279>, October 1993. DOI: [10.2307/2096279](https://doi.org/10.2307/2096279)
- Krueger, A. B. and Malečková, J., Education, Poverty and Terrorism: Is There a Causal Connection?, *The Journal of Economic Perspectives*, vol. **17**, no. 4, pp. 119–44, accessed November 24, 2025, from <https://www.jstor.org/stable/3216934>, 2003.
- Lipset, S. M., Some Social Requisites of Democracy: Economic Development and Political Legitimacy, *American Political Science Review*, vol. **53**, no. 1, pp. 69–105, 1959. DOI: [10.2307/1951731](https://doi.org/10.2307/1951731)
- Lubke, G. H. and Muthén, B. O., Applying Multigroup Confirmatory Factor Models for Continuous Outcomes to Likert Scale Data Complicates Meaningful Group Comparisons, *Structural Equation Modeling: A Multidisciplinary Journal*, vol. **11**, no. 4, pp. 514–34, accessed November 24, 2025, October 2004. DOI: [10.1207/s15328007sem1104_2](https://doi.org/10.1207/s15328007sem1104_2)
- Lüdtke, D., Ggeffects: Tidy Data Frames of Marginal Effects from Regression Models, *Journal of Open Source Software*, vol. **3**, no. 26, p. 772, accessed November 24, 2025, June 2018. DOI: [10.21105/joss.00772](https://doi.org/10.21105/joss.00772)
- Marx, G. T., *Policing Protest*, University of Minnesota Press, accessed November 24, 2025, from <https://www.jstor.org/stable/10.5749/j.ctttv1tv>, 1998.
- McAdam, D. and Brandt, C., Assessing the Effects of Voluntary Youth Service: The Case of Teach for America, *Social Forces*, vol. **88**, no. 2, pp. 945–69, accessed November 24, 2025, December 2009. DOI: [10.1353/sof.0.0279](https://doi.org/10.1353/sof.0.0279)
- McAdam, Doug, The Biographical Consequences of Activism, *American Sociological Review*, vol. **54**, no. 5, p. 744, accessed November 24, 2025, from <https://www.jstor.org/stable/2117751>, October 1989. DOI: [10.2307/2117751](https://doi.org/10.2307/2117751)
- Muller, E. N. and Weede, E., Cross-National Variation in Political Violence: A Rational Action Approach, *Journal of Conflict Resolution*, vol. **34**, no. 4, pp. 624–51, accessed November 24, 2025, December 1990. DOI: [10.1177/0022002790034004003](https://doi.org/10.1177/0022002790034004003)
- Muñoz, J. and Anduiza, E., “If a Fight Starts, Watch the Crowd”: The Effect of Violence on Popular Support for Social Movements, *Journal of Peace Research*, vol. **56**, no. 4, pp. 485–98, accessed November 24, 2025, July 2019. DOI: [10.1177/0022343318820575](https://doi.org/10.1177/0022343318820575)
- Nie, N. H., Junn, J. and Stehlik-Barry, K., *Education and Democratic Citizenship in America*, Chicago: University of Chicago Press, 1996.
- Norman, G., Likert Scales, Levels of Measurement and the “Laws” of Statistics, *Advances in Health Sciences Education*, vol. **15**, no. 5, pp. 625–32, accessed November 24, 2025, December 2010. DOI: [10.1007/s10459-010-9222-y](https://doi.org/10.1007/s10459-010-9222-y)
- Norris, P., *Democratic Deficit: Critical Citizens Revisited*, Cambridge University Press, accessed June 24, 2026, 2011.
- Olcese, C., Saunders, C. and Tzavidis, N., In the Streets with a Degree: How Political Generations, Educational Attainment

- and Student Status Affect Engagement in Protest Politics, *International Sociology*, vol. **29**, no. 6, pp. 525–45, accessed April 8, 2026, November 2014. DOI: [10.1177/0268580914551305](https://doi.org/10.1177/0268580914551305)
- Passy, F. and Monsch, G.-A., *Biographical Consequences of Activism*, in *The Wiley Blackwell Companion to Social Movements*, D. A. Snow S. A. Soule H. Kriesi and H. J. McCammon, Eds., Wiley, accessed November 24, 2025, pp. 499–514, 2018.
- Pérez-Liñán, A. S. and Pérez-Liñán, A. S., *Presidential Impeachment and the New Political Instability in Latin America*, New York, NY: Cambridge University Press, 2010.
- Pleyers, G., El Estallido Chileno a La Luz de La Década Global de Los Movimientos Sociales, *Polis*, vol. **22**, no. 65, May 2023. DOI: [http://dx.doi.org/10.32735/s0718-6568/2023-n65-1868](https://dx.doi.org/10.32735/s0718-6568/2023-n65-1868)
- Polletta, F. and Jasper, J. M., Collective Identity and Social Movements, *Annual Review of Sociology*, vol. **27**, no. 1, pp. 283–305, accessed November 24, 2025, August 2001. DOI: [10.1146/annurev.soc.27.1.283](https://doi.org/10.1146/annurev.soc.27.1.283)
- Raudenbush, S. and Bryk, A. S., *Hierarchical Linear Models: Applications and Data Analysis Methods*, Thousand Oaks, Calif.: Sage Publ, 2010.
- Rivera, S., Severino, F. and Visconti, G., Exposure to Protests and Support for Different Forms of Violence: Evidence from the 2019 Social Outburst in Chile, *Research & Politics*, vol. **12**, no. 3, p. 20531680251355242, accessed November 24, 2025, July 2025. DOI: [10.1177/20531680251355242](https://doi.org/10.1177/20531680251355242)
- Rodríguez, Á., Peña, S., Cavieres, I., Vergara, M. J., Pérez, M., Campos, M., Peredo, D., et al., Ocular Trauma by Kinetic Impact Projectiles During Civil Unrest in Chile, *Eye*, vol. **35**, no. 6, pp. 1666–72, accessed November 24, 2025, June 2021. DOI: [10.1038/s41433-020-01146-w](https://doi.org/10.1038/s41433-020-01146-w)
- Sabucedo, J.-M., Dono, M., Alzate, M. and Seoane, G., The Importance of Protesters' Morals: Moral Obligation as a Key Variable to Understand Collective Action, *Frontiers in Psychology*, vol. **Volume 9 - 2018**, 2018.
- Siegel, H., Kohlberg, Moral Adequacy, and the Justification of Educational Interventions, *Educational Theory*, vol. **31**, no. 3–4, pp. 275–84, accessed November 24, 2025, October 1981. DOI: [10.1111/j.1741-5446.1981.tb00970.x](https://doi.org/10.1111/j.1741-5446.1981.tb00970.x)
- Singer, J. D. and Willett, J. B., *Applied Longitudinal Data Analysis: Modeling Change and Event Occurrence*, Oxford University Press New York, accessed November 24, 2025, 2003.
- Snijders, T. A. B. and Bosker, R. J., *Multilevel Analysis: An Introduction to Basic and Advanced Multilevel Modeling*, Los Angeles, Calif.: SAGE, 2012.
- Snow, D. A. and Bedford, R. D., Ideology, Frame Resonance, and Participant Mobilization, *International Social Movement Research*, vol. **1**, pp. 197–218, 1988.
- Somma, N. M., Bargsted, M., Disi Pavlic, R. and Medel, R. M., No Water in the Oasis: The Chilean Spring of 2019–2020, *Social Movement Studies*, vol. **20**, no. 4, pp. 495–502, accessed November 24, 2025, July 2021. DOI: [10.1080/14742837.2020.1727737](https://doi.org/10.1080/14742837.2020.1727737)
- Steinert-Threlkeld, Z. C., Chan, A. M. and Joo, J., How State and Protester Violence Affect Protest Dynamics, *The Journal of Politics*, vol. **84**, no. 2, pp. 798–813, accessed April 8, 2026, April 2022. DOI: [10.1086/715600](https://doi.org/10.1086/715600)
- Sunshine, J. and Tyler, T. R., The Role of Procedural Justice and Legitimacy in Shaping Public Support for Policing, *Law & Society Review*, vol. **37**, no. 3, pp. 513–48, accessed November 24, 2025, from <https://www.jstor.org/stable/1555077>, 2003.
- Tarrow, S. G. Ed., *Power in Movement: Social Movements and Contentious Politics*, Cambridge [England] New York: Cam-

- bridge University Press, 1998.
- Tetlock, P. E., A Value Pluralism Model of Ideological Reasoning., *Journal of Personality and Social Psychology*, vol. **50**, no. 4, pp. 819–27, accessed November 24, 2025, April 1986. DOI: [10.1037/0022-3514.50.4.819](https://doi.org/10.1037/0022-3514.50.4.819)
- Thaler, K. M., Mueller, L. and Mosinger, E., Framing Police Violence: Repression, Reform, and the Power of History in Chile, *The Journal of Politics*, vol. **85**, no. 4, pp. 1198–1213, accessed November 24, 2025, October 2023. DOI: [10.1086/724967](https://doi.org/10.1086/724967)
- Tilly, C., *The Politics of Collective Violence*, Cambridge University Press, accessed November 24, 2025, 2003.
- Tyler, T. R., Restorative Justice and Procedural Justice: Dealing with Rule Breaking, *Journal of Social Issues*, vol. **62**, no. 2, pp. 307–26, accessed November 24, 2025, June 2006. DOI: [10.1111/j.1540-4560.2006.00452.x](https://doi.org/10.1111/j.1540-4560.2006.00452.x)
- Tyler, T. R. and Blader, S. L., The Group Engagement Model: Procedural Justice, Social Identity, and Cooperative Behavior, *Personality and Social Psychology Review*, vol. **7**, no. 4, pp. 349–61, accessed November 24, 2025, November 2003. DOI: [10.1207/S15327957PSPR0704_07](https://doi.org/10.1207/S15327957PSPR0704_07)
- Van Stekelenburg, J. and Klandermans, B., The Social Psychology of Protest, *Current Sociology*, vol. **61**, no. 5–6, pp. 886–905, accessed November 24, 2025, September 2013. DOI: [10.1177/0011392113479314](https://doi.org/10.1177/0011392113479314)
- Varlık, S., Akpınar, S., Akpınar, Ö. and Görünü, R. M., Education as an Antidote to Violence and Social Inequality, *Humanities and Social Sciences Communications*, accessed November 24, 2025, November 2025. DOI: [10.1057/s41599-025-06192-x](https://doi.org/10.1057/s41599-025-06192-x)
- Verba, S., Schlozman, K. L. and Brady, H. E., *Voice and Equality: Civic Voluntarism in American Politics*, Cambridge, Mass.: Harvard Univ. Press, 2002.
- Walgrave, S. and Wouters, R., The Missing Link in the Diffusion of Protest: Asking Others, *American Journal of Sociology*, vol. **119**, no. 6, pp. 1670–1709, accessed November 24, 2025, from <https://www.jstor.org/stable/10.1086/676853>, 2014. DOI: [10.1086/676853](https://doi.org/10.1086/676853)
- Weakliem, D. L., The Effects of Education on Political Opinions: An International Study, *International Journal of Public Opinion Research*, vol. **14**, no. 2, pp. 141–57, accessed November 24, 2025, June 2002. DOI: [10.1093/ijpor/14.2.141](https://doi.org/10.1093/ijpor/14.2.141)
- Wickham, H., *Ggplot2*, Cham: Springer International Publishing, accessed June 24, 2026, 2016.